

TUOLE, China

WMO: 526330

Lat: 38.8087N

Lon: 98.4179E

Elev: 3370

StdP: 66.87

Time Zone: 8.00 (E08)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.9	(c) -26.7	(d) -38.2	(e) 0.1	(f) -20.5	(g) -36.8	(h) 0.2	(i) -18.9	(j) 14.7	(k) -9.8	(l) 13.0	(m) -7.0	(n) 0.2	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.9	(c) 20.4	(d) 9.0	(e) 18.6	(f) 8.2	(g) 17.0	(h) 7.5	(i) 11.0	(j) 16.9	(k) 9.8	(l) 15.7	(m) 8.9	(n) 14.7	(o) 2.7	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	10.9	12.5	7.6	9.9	11.6	6.4	9.1	10.8	42.4	17.1	38.6	15.8	36.0	14.7	16.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.8	(b) 10.3	(c) 8.7	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 12.8	(b) 10.3	(c) 8.7	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.9	-16.5	-12.6	-7.4	-1.4	5.0	8.3	10.7	9.7	5.2	-1.4	-8.7	-14.1		
	DBStd	9.88	3.58	3.93	3.79	3.28	3.26	2.75	2.34	2.83	2.38	3.05	3.35	3.32		
	HDD10.0	4419	820	632	538	343	157	65	17	39	146	353	562	747		
	HDD18.3	7372	1078	866	797	593	412	301	236	268	394	612	812	1005		
	CDD10.0	88	0	0	0	0	3	13	40	30	2	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	2	0	0	0	0	0	0	1	2	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	2.3	1.4	2.1	3.1	3.4	3.1	2.7	2.6	2.3	2.1	1.7	1.6	1.3		
(15) Precipitation	PrecAvg	284	1	2	4	9	35	62	77	59	28	7	1	0		
	PrecMax	380	4	9	12	27	86	130	132	100	74	18	5	2		
	PrecMin	203	0	0	0	0	0	24	37	22	0	0	0	0		
	PrecStd	46	1	2	3	8	18	26	24	20	17	5	2	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.3	4.6	7.9	15.0	18.7	20.0	23.2	22.7	18.6	12.6	5.9	-0.2		
		MCWB	-7.0	-3.4	-1.7	2.4	5.7	7.4	11.3	9.6	9.8	1.8	-2.1	-6.0		
	2%	DB	-4.4	1.0	5.6	11.5	16.3	18.3	20.6	20.3	15.2	9.8	3.2	-2.7		
		MCWB	-9.0	-5.4	-2.8	0.5	4.8	6.9	9.9	8.9	5.6	0.4	-3.7	-7.8		
	5%	DB	-6.4	-1.5	3.4	8.8	14.0	16.7	18.6	18.4	13.4	7.9	1.3	-4.5		
		MCWB	-10.5	-6.9	-3.7	-0.8	3.5	6.4	8.9	8.4	4.5	-0.3	-5.0	-8.9		
	10%	DB	-8.4	-3.5	1.3	6.6	12.1	15.0	16.7	16.5	11.9	6.0	-0.7	-6.4		
		MCWB	-11.8	-8.3	-4.8	-1.7	2.6	5.8	8.0	8.1	3.7	-1.3	-6.0	-10.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.8	-3.2	-1.2	2.9	7.1	9.7	12.7	12.9	11.6	2.5	-1.7	-5.9		
		MCDB	-1.6	4.1	5.7	13.6	17.2	16.1	20.4	16.6	15.8	10.5	5.3	-0.6		
	2%	WB	-9.0	-5.3	-2.3	1.0	5.5	8.6	10.9	11.1	7.6	1.1	-3.5	-7.5		
		MCDB	-4.8	0.7	4.7	9.9	13.7	15.1	17.5	16.4	12.0	8.7	2.8	-2.9		
	5%	WB	-10.4	-6.7	-3.3	-0.3	4.5	7.6	9.8	9.9	6.0	0.1	-4.7	-8.8		
		MCDB	-6.7	-1.7	3.0	7.7	12.1	14.2	15.9	15.5	11.1	6.1	0.9	-4.6		
	10%	WB	-11.8	-8.2	-4.6	-1.4	3.5	6.7	9.0	8.9	4.8	-0.8	-5.9	-10.2		
		MCDB	-8.5	-3.7	1.0	5.6	10.3	12.9	15.1	14.4	10.1	4.9	-0.8	-6.6		
(35) Mean Daily Temperature Range	MDBR		17.1	17.4	16.4	13.5	13.0	11.3	10.9	12.1	14.1	15.3	17.0	17.0		
	5% DB	MCDBR	19.2	19.4	19.0	17.5	15.5	14.6	14.3	15.5	16.8	17.3	18.8	18.0		
		MCWBR	15.3	14.4	12.7	8.7	6.7	5.9	5.8	6.0	7.9	8.7	12.6	13.8		
	5% WB	MCDBR	18.9	18.7	17.8	15.0	13.0	11.5	11.9	10.9	13.2	15.3	18.0	17.7		
		MCWBR	15.2	13.8	11.8	8.1	6.0	5.1	5.5	5.2	7.0	8.1	12.0	13.7		
(40) Clear-Sky Solar Irradiance	taub		0.223	0.262	0.323	0.383	0.407	0.406	0.386	0.374	0.335	0.289	0.242	0.215		
	taud		2.453	2.327	2.116	1.969	1.976	2.069	2.218	2.242	2.310	2.343	2.436	2.488		
	Ebn at Noon		985	972	930	890	873	872	887	890	908	924	948	975		
	Edh at Noon		78	103	142	177	180	164	140	133	116	100	79	70		
(44) All-Sky Solar Radiation	RadAvg		2.84	3.69	4.93	5.90	6.29	6.28	6.11	5.71	4.93	4.16	3.13	2.62		
	RadStd		0.17	0.21	0.31	0.32	0.34	0.41	0.39	0.39	0.30	0.22	0.17	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+99	+45

Nomenclature: See separate page